



MINING DEALER BEST PRACTICE

Salvage Jobs Standardization

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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February 2016
1506-4.02-1380

1.0 Introduction

Borusan Makina Kazakhstan has implemented Repair Drawing in the Machine Shop to standardize salvage operations.

Previously, every time the component came in for salvage, the machine operator spent a lot of time searching for information in SIS (for specifications), plus send a request for the drawing required and wait for it. Besides, it created a number of problems like unclear repair requirements, lack of quality control, no repair history, no salvage time planning.

The implementation of Repair drawing has reduced pre-salvage time, standardized salvage operations, improved the quality control and allowed to keep the repair history and plan the salvage operation time.

2.0 Best Practice Description

Repair Drawing combines component salvage specifications and drawing in one document. It consists of four main parts:

- I - **General information about the component with Repair Drawing No**, i.e. W/O No, Component name, Machine Model, Part No and Qty.
- II - **Component salvage specification** with the time allocated per each operation, i.e. Operation No, Damage Description, Dimensions before Salvage, Standard Operation Name & Description (or how the defect will be fixed), Material, Procedure & Tools (recommended to use), Technician Name, Planned Time, Actual Time.
- III - **Quality Control field** with the place for sticker of **QC Passed**,
- IV - **Drawing**.

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BORUSAN CAT		Repair drawing			#00 Page 1 of 2																																																										
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Page 1. Repair Drawing

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Part IV						

Page 2. Repair Drawing

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Repair Drawings are currently kept in MS Word. In the future within the planned implementation of CDOMS (Component Daily Operations Monitoring System) in the Machine Shop, they will be transferred to the software for a more convenient application.

3.0 Implementation Steps

In order to prepare Repair Drawing the following steps should be done:

- Determine the defect of the component (usually it is done during the inspection)
- Find the procedure that will fix the defect in the SIS
- Divide by operations in the Repair Drawing template
- Complete all columns of the Repair Drawing template except the columns indicating dimensions before salvage, technician name and actual time spent for each operations
- Attach the drawing that was taken from the SIS or DSN inquiry
- Give the ready Repair Drawing the operator (the operator will complete the rest columns)
- Collect the Repair Drawing after the salvage operations is finished
- Keep the Repair Drawing for the Repair History

4.0 Benefits

- Reduced pre-salvage time (there is no need to search for specs and drawings)
- Standardized salvage / repair operations
- Improved Quality Control
- Repair History are kept
- Improved planning time for component salvage

5.0 Resources Required

- A Qualified Service Engineer with enough engineering experience and a SIS user.
- Laptop & printer with A4 Paper.
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6.0 Supporting Attachments / References

EDITOR'S NOTE: Upon review, Caterpillar has the following recommendations to go with this Best Practice:

- Recommend establishing some sort of dating system on the prints, so they know when the print was current.
- Recommend establishing a print maintenance program, with the technical svc's group, to assure any changes are installed into their prints, as they are announced through Svc Magazines, or other update publications.
- This would parallel any other product update maintenance programs the dealer has for their major component rebuild control documents.
- Recommend dealer building their salvage standard jobs with their traditional standard job system (that they use for rebuilding major components) if possible, instead of this Best Practice (different) system.

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7.0 Related Best Practices

N/A

8.0 Acknowledgements

The idea of creating & implementing this Best Practice belongs to:

Yenlik MIRZABEKOVA
Machine Shop Service Engineer
Borusan Makina Kazakhstan LLP
ymirzabekova@borusan.com
+7 (701) 763-97-87

Yevgeniy MARSHAL
Machine Shop Team Leader
Borusan Makina Kazakhstan LLP
ymarshal@borusan.com
+ 7 (701) 461 44 76

This Best Practice was written by:
Gulnaz MANSUROVA
CRC Process Engineer
Borusan Makina Kazakhstan LLP
gmansurova@borusan.com
+7 (701) 951 98 14

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